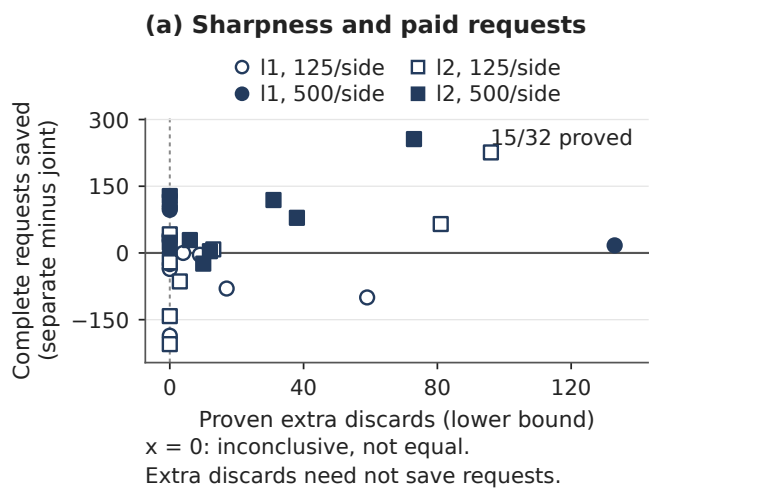
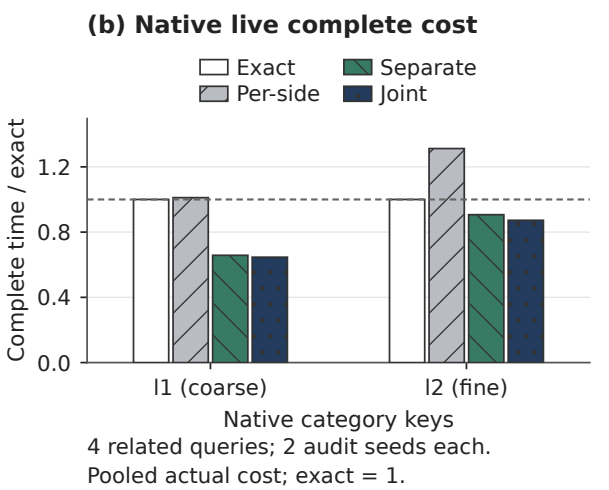

(a) Sharpness and paid requests
l1, 125/side
l2, 125/side
l1, 500/side
l2, 500/side
15/32 proved
Complete requests saved
(separate minus joint)
300
150
0
−150
0
40
80
120
Proven extra discards (lower bound)
x = 0: inconclusive, not equal.
Extra discards need not save requests.


(b) Native live complete cost
Exact
Separate
Per-side
Joint
Complete time / exact
1.2
0.8
0.4
0.0
l1 (coarse)
l2 (fine)
Native category keys
4 related queries; 2 audit seeds each.
Pooled actual cost; exact = 1.